AI-Integrated ERP

Project at a glance – where we start

Handover of the previous group's work
Repo: AI-Integrated-ERP · current version: **V3**

What it is: a chat assistant for the **Odoo** ERP. A **local AI model** (Llama 3.1 via Ollama) understands a request such as *"sales order for Pepsi, 2 chairs"*, a human clicks **approve**, and the record is created in Odoo. Intended principle: **assist, don't automate blindly.**

HOW ONE REQUEST FLOWS (V3 – VERIFIED BY RUNNING IT)

1 · Chat UI — React + Vite streams answers
›
2 · API — FastAPI keeps chat history
›
3 · AI model — Llama 3.1 picks a tool: read or create
›
4 · Check — names → Odoo IDs; unknown name = reject
›
5 · Human OK — approve / decline (creates only)
›
6 · Odoo — XML-RPC → draft order / invoice

Then the AI is called a **second time** to write a friendly summary. Two AI calls in a row for every message.

HOW IT EVOLVED

V1
May · one script

- Proved the AI can call a function
- Read-only
- Password typed in the code

V2
May–Jun · terminal chat

- Adds create + invoices
- Teams alerts
- Many "repair the AI output" hacks

V3 CURRENT
Jun · full app

- Web chat + backend API
- Name → ID lookup
- Approve / decline dialog

WHERE IT REALLY STANDS (TESTED ON ODOO 19 COMMUNITY)

Works

- ✔ Login to Odoo, read any data
- ✔ Creates a **draft** sales order and **draft** invoice after approval
- ✔ Declining creates nothing
- ✔ Invented customer names are rejected
- ✔ Streaming chat UI builds and runs

Needs fixing

- ! **2+ users at once break it** (shared connection)
- ! **Abandoned approvals freeze** the server
- ! Fuzzy name match picks the **wrong** customer/product silently
- ! AI-supplied **prices are trusted** (10 instead of 50)
- ! Can read **any** Odoo data; no tests, no Docker
- ! Orders/invoices **not linked**; nothing confirmed or posted

KEY NUMBERS

43	~1,100	28 / 40	8–200 ms	2
commits Feb–Jun 2026	lines of Python (V1–V3; V3 ≈ 600)	parallel requests failed in my test	Odoo response (not the bottleneck)	AI calls per message → likely cause of slowness

Not tested: Llama 3.1's real accuracy and speed (model download blocked in my test environment – a scripted stand-in was used). Previous group's Odoo version/hardware unknown. Security: an Odoo password and a Teams webhook are stored in the old code/history → revoke them.

The Plan

Trustworthy AI invoicing – fast, checked, measured

Starts week 7 · weekly 30-min supervisor session
Live demo (or video) · freeze 2 weeks before

> **Goal:** a chat that turns *"invoice Pepsi for 3 chairs"* into a **correct, validated, human-approved draft invoice** in Odoo – **fast**, with **measured accuracy and measured speed.**

TARGET FLOW: THE AI ONLY EXTRACTS, CODE DOES THE CHECKING

1. **Understand** – AI returns **structured** data: customer, lines, date
2. **Validate** – real customer/product, price list, duplicates, "did you mean...?"
3. **Approve** – card shows lines, totals, tax, warnings
4. **Draft in Odoo** – linked to order; confirm / post only on approval
5. **Audit** – log of who approved what

FOUR WORKSTREAMS (NOT YET ASSIGNED TO PEOPLE)

1 · Backend & Odoo `code`

- Fix concurrency + approval freeze
- Validation rules, invoice action
- Audit log, Docker, remove secrets

2 · AI, speed & UI `code`

- Structured output, slimmer prompts
- Timing, 1 AI call, compare models
- Approval screen, test runner

3 · Product & Odoo domain `no code`

- Previous group, supervisor, backlog
- Odoo test data, business rules
- Demo story, "Ask AI" check

4 · Quality, research & docs `no code`

- 50-prompt test set, manual testing
- Experiment log (grading evidence)
- Docs, slides, backup video

TIMELINE

- **Before wk 7** – Email previous group · set up Odoo + AI · revoke secrets · baseline speed
- **Sprint 1** – Week 7 · safe & measurable: fixes, timings, test set v1
- **Sprint 2** – Week 8 · invoice pipeline v1: before/after numbers
- **Sprints 3–4** – Link order↔invoice · audit · model benchmark · approval card
- **Sprint 5+** – Harden; decide stretch idea
- **Freeze** – 2 weeks before end · bug fixes only · record video
- **Demo** – Live demo + backup video

TOP 5 THINGS TO FIX FIRST

1. **Concurrency** – one Odoo connection per request
2. **Approval freeze** – pending actions that expire
3. **Validation layer** – ambiguity, prices, duplicates, allow-list
4. **Latency** – measure, 1 AI call, keep model loaded, shorter history
5. **Test set** – 50 prompts with expected results

Do immediately: revoke the leaked Odoo password and Teams webhook.

WHY THIS IS NEW

Odoo's own **Ask AI** can't change data, needs OpenAI/Gemini or paid credits, and has **no local-model option**. Our edge: **Odoo Community + a local model**, with **checked, approved, logged** writes and **published accuracy and speed numbers**.

Stretch only: local PDF/e-mail → draft invoice.